

UNIT OF COMPETENCE	Network design & management
UNIT CODE	0612 451 07A
NAME OF THE TRAINER	JAMES NDIRANGU
COMPETENCE LEVEL	LEVEL 5
INSTITUTION	THE ABERDARE TECHNICAL COLLEGE
Number of learners	3
DATE OF PREPARATION	06/08/2026

SKILL OR JOB TASK

Designing computer network, installing computer network, testing computer network and performing computer network maintenance.

BENCHMARK OR CRITERIA TO BE USED
1. Design computer network <i>1.1 User needs are collected as per the workplace procedure.</i> <i>1.2 Physical network design is developed as per user requirements.</i>

1.3 Logical network design is developed as per user requirements.

1.4 Computer network design is mapped out as per user requirements.

2. Install computer network

2.1 Safety measures are observed as per workplace procedure.

2.2 Computer network components are identified as per the network design.

2.3 Computer network is set up as per the network design.

2.4 Computer network is configured as per the network design.

2.5 Computer network is documented as per the network design.

2.6 Computer network comp

3. Test computer network

3.1 Network testing tools and equipment are identified as per the work requirement.

3.2 Network components are tested as per the workplace procedure.

3.3 Network testing report is developed as per the work requirement.

4. Perform computer network maintenance.

4.1 Computer network maintenance schedule is prepared as per workplace procedure.

4.2 Computer network is monitored as per maintenance schedule.

4.3 Computer network is optimized as per the user requirements.

4.4 Computer network maintenance report is developed as per workplace procedure.

LEARNING OUTCOME 1: DESIGN COMPUTER NETWORK

WEEK 1

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
1	1. Design Computer Network	Session 1: User Needs Collection • Establishing user needs • Identification of user needs • Analysis of user needs • Documentation of user needs Session 2: Introduction to Computer Networking • Definition of computer network •	By the end of this lesson the trainees should be able to: 1. Identify user needs for a computer network. 2. Analyse identified user requirements. 3. Document network user	1. Explain the process of collecting user needs. 2. Demonstrate techniques for analysing user requirements. 3. Demonstrate documentation of user requirements. 4. Define computer networking. 5. Explain	1. Identify network user needs. 2. Analyse sample user requirements. 3. Document the identified requirements. 4. Define a computer network. 5. Explain networking terms. 6. Illustrate basic	Computers, projector, networking notes, user-requirement forms, charts	Oral questions, group discussion, user-needs exercise	Date: _____ Reflection: _____

		Computer networking terms • Basic networking concepts	requirements. 4. Define a computer network. 5. Explain common computer networking terms. 6. Illustrate basic networking concepts.	common networking terminology. 6. Illustrate basic networking concepts using examples.	networking concepts.			
1	1. Design Computer Network	Session 3: Types of Computer Networks • LAN • WAN • PAN • MAN • Characteristics • Applications • Advantages and limitations	By the end of this lesson the trainees should be able to: 1. Identify different types of computer networks. 2. Describe characteristics of LAN, WAN, PAN and MAN. 3. Differentiate LAN, WAN, PAN and MAN. 4.	1. Introduce the different network types. 2. Describe characteristics of each network type. 3. Compare LAN, WAN, PAN and MAN. 4. Demonstrate network classification. 5. Discuss advantages and limitations. 6.	1. Identify network types. 2. Describe their characteristics. 3. Differentiate LAN, WAN, PAN and MAN. 4. Classify networks according to coverage. 5. Compare advantages and limitations. 6. Select	Network diagrams, computers, projector, internet, notes	Written exercise, classification activity, oral questions	Date: _____ Reflection: _____

			Classify networks according to geographical coverage. 5. Compare advantages and limitations of different network types. 6. Select an appropriate network type for a given situation.	Provide scenarios for selecting suitable network types.	suitable network types for given scenarios.			
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WEEK 2

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
2	1. Design Computer Network	Session 4: Network Topologies • Star topology • Ring topology • Mesh topology •	By the end of this lesson the trainees should be able to: 1. Identify different	1. Identify and introduce the topologies. 2. Describe characteristics of each	1. Identify network topologies. 2. Describe their characteristics. 3. Differentiate	Topology charts, computers, projector, drawing materials, Cisco	Topology drawing exercise, oral questions	Date: _____ Reflection: _____

		Hybrid topology • Point-to-point topology • Characteristics • Advantages and disadvantages • Applications	network topologies. 2. Describe characteristics of each topology. 3. Differentiate the specified network topologies. 4. Compare advantages and disadvantages of each topology. 5. Select an appropriate topology for a given network requirement. 6. Draw network topology diagrams.	topology. 3. Demonstrate differences among the topologies. 4. Compare advantages and disadvantages. 5. Guide selection of suitable topologies. 6. Demonstrate topology diagrams.	e the topologies. 4. Compare their advantages and disadvantages. 5. Select appropriate topologies. 6. Draw topology diagrams.	Packet Tracer		
2	1. Design Computer Network	Session 5: Network Components • Switches • Hubs • Routers •	By the end of this lesson the trainees should be able to: 1.	1. Demonstrate different network components. 2. Explain	1. Identify network components. 2. Describe device functions. 3.	Switches, hubs, routers, computers, cables, connectors	Component identification practical	Date: _____ Reflection: _____ —

		Ports • Computers/e nd devices • Transmission media	Identify network components. 2. Describe the functions of switches, hubs and routers. 3. Explain the purpose of network ports. 4. Classify network devices according to their functions. 5. Identify different transmission media. 6. Select appropriate network components for a given network design.	functions of switches, hubs and routers. 3. Demonstrate different network ports. 4. Guide classification of network devices. 5. Demonstrate transmission media. 6. Guide selection of components.	Identify network ports. 4. Classify devices. 5. Identify transmission media. 6. Select appropriate components.	s, projector		
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WEEK 3

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
3	1. Design Computer Network	Session 6: Physical Network Design Development • Physical network design • Physical layout • Network devices and locations • Cabling paths • Data points • Access points	By the end of this lesson the trainees should be able to: 1. Explain physical network design. 2. Identify requirements for a physical network layout. 3. Select appropriate locations for network devices. 4. Develop a physical	1. Explain physical network design. 2. Demonstrate physical network layout development. 3. Demonstrate placement of network devices. 4. Guide development of physical layouts. 5. Demonstrate how user requirements are	1. Explain physical network design. 2. Identify physical layout requirements. 3. Select device locations. 4. Develop a physical network layout. 5. Incorporate user requirements into the design.	Computer s, floor plans, draw.io, projector, measuring tools	Physical network design practical	Date: _____ Reflection: _____

			network layout. 5. Apply user requirements when developing a physical design.	incorporate d.				
3	1. Design Computer Network	Session 7: Logical Network Development • Logical network design • Logical topology • IP addressing plan • Network segmentation • Logical network diagrams	By the end of this lesson the trainees should be able to: 1. Explain logical network design. 2. Analyse user requirements for logical network design. 3. Develop a logical network topology. 4. Construct a logical network diagram. 5. Prepare a	1. Explain logical network design. 2. Guide analysis of user requirements. 3. Demonstrate logical topology development. 4. Demonstrate logical network diagrams. 5. Demonstrate preparation of an IP addressing plan. 6. Guide	1. Explain logical network design. 2. Analyse user requirements. 3. Develop a logical topology. 4. Construct a logical network diagram. 5. Prepare an IP addressing plan. 6. Evaluate the logical design.	Computers, draw.io, Cisco Packet Tracer, IP addressing worksheets	Logical design practical	Date: _____ Reflection: _____

			basic IP addressing plan. 6. Evaluate a logical design against user requirements.	evaluation of logical designs.				
3	1. Design Computer Network	Session 8: Computer Network Design Methodology • Network design overview • Network design methodology • Hierarchical network design • Flat network design • Greenfield site • Brownfield site	By the end of this lesson the trainees should be able to: 1. Explain the concept of computer network design. 2. Describe the stages of network design methodology. 3. Differentiate hierarchical and flat network designs. 4. Compare	1. Explain network design concepts. 2. Describe the design methodology. 3. Compare hierarchical and flat designs. 4. Differentiate Greenfield and Brownfield sites. 5. Present scenarios for selecting suitable approaches.	1. Explain network design. 2. Describe design methodology. 3. Differentiate hierarchical and flat designs. 4. Compare Greenfield and Brownfield sites. 5. Select appropriate approaches for given scenarios.	Network design diagrams, case studies, computers, projector	Case study, group presentation	Date: _____ Reflection: _____

			Greenfield and Brownfield sites. 5. Select an appropriate design approach for a given site.					
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WEEK 4

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
4	1. Design Computer Network	Session 9: Site Preparation and Network Design Goals - Site preparation - Floor plan preparation - Data/access points - Implementation of user	By the end of this lesson the trainees should be able to: Explain network site preparation requirements. Apply site preparation procedures.	Explain site preparation requirements. Demonstrate site preparation procedures. Demonstrate floor plan development.	Explain site preparation requirements. Apply site preparation procedures. Develop a network floor plan.	Floor plans, computers, draw.io, Cisco Packet Tracer, projector, measuring tools	CAT 1: Complete network design practical	Date: _____ Reflection: _____

		needs • Scalability • Availability • Security • Manageability	3. Develop a network floor plan. 4. Determine appropriate locations for data and access points. 5. Apply user needs to a network design. 6. Evaluate a network design based on scalability, availability, security and manageability. 7. Develop a complete network design proposal.	Demonstrate placement of data and access points. 5. Guide implementation of user needs. 6. Demonstrate evaluation of design goals. 7. Guide development of the final design proposal.	Determine data and access point locations. 5. Implement user requirements in the design. 6. Evaluate the design. 7. Develop a complete network design proposal.			
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LEARNING OUTCOME 2: INSTALL COMPUTER NETWORK

WEEK 5

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
5	2. Install Computer Network	Session 10: Network Installation Safety • PPE • Overall/apron/dust coat • Antiglare screens • Dust mask • Gloves • Antistatic equipment • Ergonomics • First Aid kit • Cable management • Cable routing • Cable labelling	By the end of this lesson the trainees should be able to: 1. Identify hazards associated with network installation. 2. Explain the importance of PPE and safe working practices. 3. Select appropriate PPE for specific tasks. 4. Demonstrate correct use of PPE and	1. Identify network installation hazards. 2. Explain PPE and safe working practices. 3. Demonstrate selection of PPE. 4. Demonstrate PPE and antistatic equipment use. 5. Demonstrate ergonomic practices. 6. Demonstrate cable routing, management and labelling.	1. Identify installation hazards. 2. Explain safety practices. 3. Select suitable PPE. 4. Use PPE and antistatic equipment. 5. Apply ergonomic practices. 6. Apply cable routing, management and labelling procedures.	PPE, antistatic wrist straps, gloves, dust masks, first aid kit, cable ties, labels	Safety observation checklist	Date: _____ Reflection: _____ _____

			antistatic equipment. 5. Apply ergonomic practices during network installation. 6. Apply appropriate cable routing, management and labelling procedures.					
5	2. Install Computer Network	Session 11: Electrical, Fire and Emergency Safety • Electrical safety • • Insulated tools • • Power ratings • Fire safety • Classes of fire • Fire extinguishers • Emergency procedures • Emergency contacts • Contingency procedures	By the end of this lesson the trainees should be able to: 1. Identify electrical and fire hazards. 2. Explain electrical safety requirements. 3. Select appropriate insulated	1. Identify electrical and fire hazards. 2. Explain electrical safety. 3. Demonstrate use of insulated tools. 4. Explain power ratings. 5. Demonstrate fire classificatio	1. Identify hazards. 2. Explain electrical safety. 3. Select insulated tools. 4. Interpret power ratings. 5. Classify fires. 6. Select appropriate extinguishe	Fire extinguishers, first aid kit, insulated tools, safety charts, PPE	Practical safety demonstration	Date: _____ Reflection: _____ _____

			tools for electrical tasks. 4. Interpret equipment power ratings. 5. Classify different classes of fire. 6. Select appropriate fire extinguishers. 7. Apply emergency and contingency procedures.	n. 6. Demonstrate appropriate fire extinguisher selection. 7. Demonstrate emergency procedures.	emergency procedures.			
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WEEK 6

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
6	2. Install Computer	Session 12: Identification of Network Components •	By the end of this lesson the trainees	1. Demonstrate network component	1. Identify network components. 2.	Switches, routers, APs, cables, connectors,	Component identification practical	Date: _____ Reflection:

	Network	Switches • Routers • Transmission media • Connectors • Wireless access points • Networking software and management tools • Security devices • Servers • Storage devices	should be able to: 1. Identify network installation component s. 2. Describe the functions of switches and routers. 3. Differentiate network transmission media and connectors. 4. Identify wireless access points. 5. Classify networking software and management tools. 6. Identify network security devices. 7.	s. 2. Explain switch and router functions. 3. Demonstrate transmission media and connectors. 4. Demonstrate wireless access points. 5. Explain networking software and management tools. 6. Demonstrate security devices. 7. Explain server and storage requirements.	Describe switch and router functions. 3. Differentiate transmission media and connectors. 4. Identify wireless APs. 5. Classify networking software and tools. 6. Identify security devices. 7. Select suitable servers and storage devices.	servers, storage devices, software		
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			Select appropriate servers and storage devices.					
6	2. Install Computer Network	Session 13: Network Installation Tools • Crimpers • Cable strippers • Cutters/scissors • Screwdrivers • Pliers • Cable tie tools • Fibre tools • Insertion/extraction tools • Manual/automatic switch boxes • Network testers • Punch-down tools • Driver installers • Multimeter • Tone generator and probe	By the end of this lesson the trainees should be able to: 1. Identify network installation tools. 2. Describe the functions of each tool. 3. Select appropriate tools for specific installation tasks. 4. Demonstrate safe use of cable preparation and termination tools. 5. Demonstrate	1. Display and identify network tools. 2. Explain tool functions. 3. Demonstrate selection of tools. 4. Demonstrate cable preparation tools. 5. Demonstrate testing tools. 6. Demonstrate tool safety procedures.	1. Identify network tools. 2. Describe tool functions. 3. Select suitable tools. 4. Use cable preparation tools safely. 5. Use testing tools safely. 6. Apply tool safety procedures.	Crimpers, strippers, cutters, screwdrivers, pliers, fibre tools, punch-down tools, testers, multimeter, tone generator	Tool identification and practical demonstration	Date: _____ Reflection: _____

			te safe use of testing tools. 6. Apply appropriate tool safety procedures.					
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WEEK 7

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
7	2. Install Computer Network	Session 14: Network Installation Materials and Cabling • Network cables • Trunking covers • Connectors • RJ45 sockets • Patch cords • Cable ties	By the end of this lesson the trainees should be able to: 1. Identify network installation materials. 2. Describe the uses of network cables and	1. Display network installation materials. 2. Explain their uses. 3. Guide material selection. 4. Demonstrate interpretation of network layouts. 5.	1. Identify installation materials. 2. Describe their uses. 3. Select suitable materials. 4. Interpret network layouts. 5. Apply cabling standards. 6.	Cat5e/Cat 6 cables, trunking, RJ45 sockets, patch cords, cable ties, floor plans	Material identification and installation planning	Date: _____ Reflection: _____ -

		Network layout • Cabling standards and codes • Cable routing	accessories. 3. Select appropriate materials for an installation. 4. Interpret a network layout. 5. Apply cabling standards and codes. 6. Plan appropriate cable routes.	Explain cabling standards. 6. Demonstrate cable routing.	Plan cable routes.			
7	2. Install Computer Network	Session 15: Cable Termination and Installation • T568A colour code • T568B colour code • RJ45 termination • Cable installation • Cable management • Cable labelling •	By the end of this lesson the trainees should be able to: 1. Explain T568A and T568B cabling standards. 2. Differentiate T568A and T568B colour codes. 3. Apply the appropriate	1. Explain T568A and T568B. 2. Demonstrate the colour codes. 3. Demonstrate application of the standards. 4. Demonstrate RJ45 termination. 5. Demonstrate cable installation.	1. Explain T568A and T568B. 2. Differentiate the colour codes. 3. Apply the correct standard. 4. Terminate RJ45 cables. 5. Install network cables. 6. Manage and label cables. 7. Test	Crimper, cable stripper, RJ45 connectors, Cat5e/Cat 6 cables, cable tester, trunking, cable ties	Practical Assessment 1: Cable termination, installation and testing	Date: _____ Reflection: _____

		Testing installed cables	colour code when terminating cables. 4. Terminate RJ45 cables correctly. 5. Install network cables according to a design. 6. Organize and label installed cables. 7. Test installed cables for continuity and correct termination.	6. Demonstrate cable management and labelling. 7. Demonstrate cable testing.	installed cables.			
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WEEK 8

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates

8	2. Install Computer Network	Session 16: Network Models and IP Addressing • OSI model • TCP/IP model • IPv4 • IPv6 • IP address classes • Classful addressing • CIDR • Subnetting • Static addressing • Dynamic addressing	By the end of this lesson the trainees should be able to: 1. Identify layers of the OSI and TCP/IP models. 2. Explain the functions of the layers. 3. Differentiate IPv4 and IPv6. 4. Classify IPv4 addresses using address classes. 5. Calculate CIDR and subnetting requirements. 6. Apply static and dynamic IP addressing.	1. Explain OSI and TCP/IP models. 2. Demonstrate layer functions. 3. Compare IPv4 and IPv6. 4. Demonstrate IPv4 address classes. 5. Demonstrate CIDR and subnetting. 6. Demonstrate static and dynamic addressing.	1. Identify model layers. 2. Explain layer functions. 3. Differentiate IPv4 and IPv6. 4. Classify IP addresses. 5. Calculate subnets. 6. Configure static and dynamic addressing.	Cisco Packet Tracer, IP worksheets, routers, switches, computers	IP addressing and subnetting practical	Date: _____ Reflection: _____
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8	2. Install Computer Network	Session 17: Switch and Router Configuration • Device setup • Interface configuration • IP assignment • Basic switch configuration • Basic router configuration • Network connectivity configuration	By the end of this lesson the trainees should be able to: 1. Identify basic switch and router configuration requirements. 2. Explain the purpose of device interfaces. 3. Configure switch interfaces. 4. Configure router interfaces. 5. Assign appropriate IP addresses to interfaces. 6.	1. Explain device configuration requirements. 2. Demonstrate network interfaces. 3. Demonstrate switch configuration. 4. Demonstrate router configuration. 5. Demonstrate IP assignment. 6. Demonstrate connectivity configuration.	1. Identify configuration requirements. 2. Explain interface functions. 3. Configure switch interfaces. 4. Configure router interfaces. 5. Assign IP addresses. 6. Establish connectivity.	Cisco Packet Tracer, switches, routers, computers, configuration guides	Device configuration practical	Date: _____ Reflection: _____
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			Establish basic network connectiv y.					
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LEARNING OUTCOME 2: INSTALL COMPUTER NETWORK – CONTINUED

WEEK 9

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
9	2. Install Computer Network	Session 18: Routing Protocols and Network Segmentation • RIP • EIGRP •	By the end of this lesson the trainees should be able to: 1. Explain the purpose of	1. Explain routing protocols. 2. Compare RIP, EIGRP and OSPF. 3. Demonstrate routing	1. Explain routing protocols. 2. Differentiate RIP, EIGRP and OSPF. 3. Configure	Cisco Packet Tracer, routers, switches, computers	Routing and VLAN practical	Date: _____ Reflection: _____

		OSPF • VLANs • VLAN configuration • STP • LACP	routing protocols. 2. Differentiate RIP, EIGRP and OSPF. 3. Configure basic routing protocols. 4. Explain the purpose of VLANs. 5. Create and configure VLANs. 6. Configure STP and LACP.	configuration. 4. Explain VLAN concepts. 5. Demonstrate VLAN creation and configuration. 6. Demonstrate STP and LACP configuration.	routing protocols. 4. Explain VLANs. 5. Create and configure VLANs. 6. Configure STP and LACP.			
9	2. Install Computer Network	Session 19: Network Access, Wireless and Security Configuration • ACLs • NAT/PAT • Port security • Wireless access points • SSID • DHCP • DNS • SMTP •	By the end of this lesson the trainees should be able to: 1. Explain common network security mechanisms. 2. Configure ACLs. 3.	1. Explain network security mechanisms. 2. Demonstrate ACL configuration. 3. Demonstrate NAT/PAT configuration. 4. Demonstrate	1. Explain security mechanisms. 2. Configure ACLs. 3. Configure NAT/PAT. 4. Apply port security. 5. Configure wireless APs, SSID	Cisco Packet Tracer, routers, switches, APs, computers, security configuration guides	CAT 2 + Practical Assessment 2: Network configuration and security	Date: _____ Reflection: _____

		Wireless security • Network privileges • Firewall/security policies • Privileged accounts	Configure NAT/PAT. 4. Apply port security. 5. Configure wireless access points, SSID and DHCP. 6. Explain the roles of DNS and SMTP in network services. 7. Apply wireless security measures. 8. Configure network privileges and privileged accounts. 9. Evaluate firewall and network security policies.	Configure port security. 5. Demonstrate wireless AP, SSID and DHCP configuration. 6. Explain DNS and SMTP. 7. Demonstrate wireless security. 8. Demonstrate privilege configuration. 9. Guide evaluation of security policies.	and DHCP. 6. Explain DNS and SMTP. 7. Apply wireless security. 8. Configure network privileges. 9. Evaluate security policies.			
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9	2. Install Computer Network	Session 20: Network Documentation and Disposal of Network Waste • Definition of network documentation • Importance of documentation • Physical documentation • Logical documentation • Configuration documentation • Identification of network waste • Classification of e-waste • Hazards of e-waste • Disposal methods • Legal and	By the end of this lesson the trainees should be able to: 1. Define network documentation. 2. Explain its importance. 3. Demonstrate different forms of documentation. 4. Demonstrate preparation of documentation. 5. Identify network waste. 6. Demonstrate e-waste classification. 7. Explain e-waste hazards. 8. Explain appropriate disposal methods. 9. Discuss	1. Define network documentation. 2. Explain its importance. 3. Differentiate documentation types. 4. Prepare network documentation. 5. Identify network waste. 6. Classify e-waste. 7. Analyse e-waste hazards. 8. Select appropriate disposal methods. 9. Evaluate disposal practices.	Documentation templates, computers, network diagrams, sample e-waste, environmental guidelines	Documentation exercise, e-waste classification exercise	Date: _____ Reflection: _____
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		environmental compliance • Waste Management Act 2022 • EMCA Act 2015 • Public Procurement and Asset Disposal Act 2015	according to its characteristics. 7. Analyse hazards associated with network e-waste. 8. Select appropriate disposal methods. 9. Evaluate disposal practices against applicable legal and environmental requirements.	applicable legal and environmental requirements.				
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LEARNING OUTCOME 3: TEST COMPUTER NETWORK

WEEK 10

Week	Main Learning	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
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	Outcome							
10	3. Test Computer Network	Session 21: Introduction to Network Testing • Introduction to network testing • Importance of testing • Testing tools and equipment • Clamp meter • Voltmeter • Cable tester • Signal tester	By the end of this lesson the trainees should be able to: 1. Define network testing. 2. Explain the importance of testing a computer network. 3. Identify network testing tools and equipment. 4. Describe the functions of testing equipment. 5. Select appropriate testing equipment for specific tasks.	1. Define network testing. 2. Explain the importance of testing. 3. Demonstrate testing tools. 4. Explain equipment functions. 5. Guide selection of appropriate testing equipment.	1. Define network testing. 2. Explain its importance. 3. Identify testing tools. 4. Describe equipment functions. 5. Select suitable testing equipment.	Cable testers, clamp meter, voltmeter, signal tester, multimeter	Tool identification and oral assessment	Date: _____ Reflection: _____
10	3. Test Computer	Session 22: Network Component Testing •	By the end of this lesson the trainees should be	1. Explain the different testing approaches. 2.	1. Explain testing approaches. 2. Differentiate	Network devices, cable tester,	Network component testing practical	Date: _____ Reflection: _____

	Network	Performance testing • Functional testing • Security testing • Testing procedures • Testing standards	able to: 1. Explain performance, functional and security testing. 2. Differentiate the different network testing approaches. 3. Apply network testing procedures. 4. Perform performance tests. 5. Perform functional tests. 6. Perform security tests. 7. Evaluate test results against specified standards.	Compare testing methods. 3. Demonstrate testing procedures. 4. Demonstrate performance testing. 5. Demonstrate functional testing. 6. Demonstrate security testing. 7. Guide evaluation of results.	testing methods. 3. Apply testing procedures. 4. Perform performance tests. 5. Perform functional tests. 6. Perform security tests. 7. Evaluate test results.	computers , testing checklists		_____
10	3. Test Computer	Session 23: Network Diagnostic Tools and Commands •	By the end of this lesson the trainees should be able to: 1.	1. Identify diagnostic tools. 2. Explain their purposes. 3.	1. Identify diagnostic tools. 2. Explain their purposes. 3.	Computers , command prompt, Wireshark, internet	Diagnostic commands practical	Date: _____ Reflection: _____

	Network	Ping • Tracert • Traceroute • IPConfig • NSLookup • Speed test • Wireshark	Identify network diagnostic commands and tools. 2. Explain the purpose of each diagnostic command. 3. Apply Ping to test connectivity. 4. Apply Tracert/Trace route to analyse network paths. 5. Use IPConfig and NSLookup to obtain network information. 6. Perform network speed tests. 7. Capture and analyse network traffic using Wireshark.	Demonstrate Ping. 4. Demonstrate Tracert/Trace route. 5. Demonstrate IPConfig and NSLookup. 6. Demonstrate speed testing. 7. Demonstrate Wireshark traffic capture and analysis.	Use Ping. 4. Use Tracert/Trace route. 5. Use IPConfig and NSLookup. 6. Perform speed tests. 7. Capture and analyse traffic using Wireshark.	connection, network devices		
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10	3. Test Computer Network	Session 24: Network Test Report • Importance of test reports • Components of a network test report • Test results • Analysis and recommendations • Presentation techniques • Interactive presentations	By the end of this lesson the trainees should be able to: 1. Explain the importance of a network test report. 2. Identify components of a network test report. 3. Organize network test results. 4. Analyse network test findings. 5. Prepare a network test report. 6. Present network test findings using appropriate techniques. 7. Defend recommendations based on test results.	1. Explain the importance of test reports. 2. Identify report components. 3. Demonstrate organization of test results. 4. Demonstrate analysis of findings. 5. Demonstrate report preparation. 6. Demonstrate presentation techniques. 7. Facilitate interactive presentations and feedback.	1. Explain the importance of test reports. 2. Identify report components. 3. Organize test results. 4. Analyse findings. 5. Prepare a test report. 6. Present test findings. 7. Defend recommendations.	Computers , report templates, projector, test data	CAT 3: Network testing and test report	
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LEARNING OUTCOME 4: PERFORM COMPUTER NETWORK MAINTENANCE

WEEK 10 – CONTINUED / FINAL MAINTENANCE SESSIONS

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainees Activities	Resources	Assessment	Reflections and Dates
10	4. Perform Computer Network Maintenance	Session 25: Network Maintenance Schedule and Troubleshooting • Importance of maintenance • Maintenance schedule • Preparation of maintenance schedule • Troubleshooting process • Troubleshooting techniques	By the end of this lesson the trainees should be able to: 1. Explain the importance of computer network maintenance. 2. Identify network maintenance requirements. 3. Prepare a network maintenance schedule. 4. Describe	1. Explain the importance of maintenance. 2. Identify maintenance requirements. 3. Demonstrate preparation of a maintenance schedule. 4. Explain the troubleshooting process. 5. Demonstrate	1. Explain maintenance importance. 2. Identify maintenance requirements. 3. Prepare a maintenance schedule. 4. Describe troubleshooting procedures. 5. Apply troubleshooting techniques. 6. Evaluate maintenance activities.	Maintenance schedules, computers, network devices, troubleshooting guides	Maintenance schedule exercise, troubleshooting practical	Date: _____ Reflection: _____

			the network troubleshooting process. 5. Apply appropriate troubleshooting techniques. 6. Evaluate maintenance activities against planned requirements.	troubleshooting techniques. 6. Guide evaluation of maintenance activities.				
10	4. Perform Computer Network Maintenance	Session 26: Network Monitoring • Ping • Tracert • NSLookup • IPConfig • Speed test • Traceroute • Wireshark • Network monitoring configuration • Performance analysis	By the end of this lesson the trainees should be able to: 1. Identify network monitoring tools. 2. Explain the purpose of network monitoring. 3. Configure	1. Identify monitoring tools. 2. Explain network monitoring. 3. Demonstrate monitoring configuration. 4. Demonstrate diagnostic commands.	1. Identify monitoring tools. 2. Explain network monitoring. 3. Configure monitoring tools. 4. Monitor network performance. 5. Capture network	Computers, Wireshark, network devices, command prompt, monitoring software	Network monitoring practical	Date: _____ Reflection: _____

			network monitoring tools. 4. Monitor network performance using diagnostic commands. 5. Capture network traffic using Wireshark. 6. Analyse network performance information. 7. Identify network performance problems from monitoring results.	5. Demonstrate Wireshark capture. 6. Demonstrate performance analysis. 7. Guide identification of performance problems.	traffic. 6. Analyse performance information. 7. Identify performance problems.			
10	4. Perform Computer Network	Session 27: Network Optimization • Network optimization techniques • Performance improvement	By the end of this lesson the trainees should be able to: 1. Explain network	1. Explain network optimization. 2. Identify performance factors. 3. Demonstrate	1. Explain optimization. 2. Identify performance factors. 3. Analyse network	Cisco Packet Tracer, network devices, computers, monitoring tools	Optimization practical exercise	Date: _____ Reflection: _____

	Maintenance	• Quality of Service (QoS) • Bandwidth management • Traffic management	optimization. 2. Identify factors affecting network performance. 3. Analyse network performance problems. 4. Apply appropriate network optimization techniques. 5. Configure basic QoS mechanisms. 6. Evaluate the effectiveness of optimization measures.	performance analysis. 4. Demonstrate optimization techniques. 5. Demonstrate QoS configuration. 6. Guide evaluation of optimization measures.	problems. 4. Apply optimization techniques. 5. Configure QoS. 6. Evaluate optimization results.			
10	4. Performance Compu	Session 28: Network Maintenance Report •	By the end of this lesson the trainees	1. Explain the importance of	1. Explain report importance. 2. Identify	Computers, report templates, projector,	Final Integrated Practical +	Date: _____ Reflection: _____

	ter Network Maintenance	Importance of maintenance reports • Components of maintenance reports • Preparation of maintenance reports • Presentation of maintenance findings	should be able to: 1. Explain the importance of network maintenance reports. 2. Identify components of a maintenance report. 3. Organize maintenance findings. 4. Analyse maintenance information. 5. Prepare a network maintenance report. 6. Present maintenance findings professionally.	maintenance reports. 2. Identify report components. 3. Demonstrate organization of findings. 4. Demonstrate analysis of maintenance information. 5. Demonstrate report preparation. 6. Guide professional presentation.	report components. 3. Organize findings. 4. Analyse maintenance information. 5. Prepare a maintenance report. 6. Present maintenance findings.	maintenance records	Maintenance Report	
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